WORKED SOLUTION

STEP I 2004, Question 9

Mechanics • Projectile motion

STEP I 2004, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q9

The key idea

The first part is the standard "vertical velocity is zero at the top" calculation. In the tunnel, the speed u isn't given, but it isn't free either: once θ is chosen, requiring the path to pass through P fixes u. So we eliminate u, write the greatest height as a function of θ alone (best in terms of $t = \tan\theta$), and ask when it stays below the roof. The answer comes out as a quadratic inequality in t.

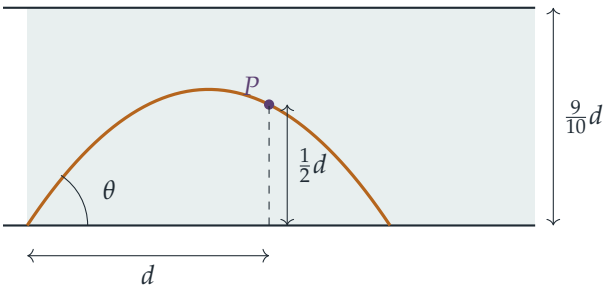
Solution

Greatest height

Model the particle as a point moving freely under gravity, with no air resistance. Take axes with origin at the point of projection, x horizontal and y vertically up. The vertical component of velocity starts at $u\sin\theta$ and has constant acceleration $-g$. At the highest point the vertical velocity is zero, so by $v^2 = v_0^2 + 2as$ in the vertical direction,

$$0 = u^2\sin^2\theta - 2gH, \qquad \text{so} \qquad H = \frac{u^2\sin^2\theta}{2g}.$$

The tunnel



The path of the particle comes from eliminating the time T between $x = uT\cos\theta$ and $y = uT\sin\theta - \frac{1}{2}gT^2$:

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = xt - \frac{gx^2(1+t^2)}{2u^2},$$

where $t = \tan\theta$ and we have used $\sec^2\theta = 1 + t^2$.

Passing through P. Putting $x = d$, $y = \frac{1}{2}d$ and dividing by d:

$$\tfrac{1}{2} = t - \frac{gd(1+t^2)}{2u^2}, \qquad \text{so} \qquad \frac{gd(1+t^2)}{2u^2} = t - \tfrac{1}{2}.$$

The left-hand side is positive, so we need $t > \frac{1}{2}$. (This makes sense: the particle starts heading in direction θ and gravity only bends the path downwards, so it can only reach P if it is aimed above the line to P, which has gradient $\frac{1}{2}$.) For each such θ there is exactly one speed that works:

$$u^2 = \frac{gd(1+t^2)}{2t-1}.$$

Ask yourself The question gives no value of u. How can the condition be on θ alone?

Because the speed is not ours to choose: the condition "passes through P" pins u down once θ is fixed. So the greatest height is really a function of θ only.

Clearing the roof. Using $\sin^2\theta = \dfrac{t^2}{1+t^2}$ in the formula for H,

$$H = \frac{u^2}{2g}\cdot\frac{t^2}{1+t^2} = \frac{d(1+t^2)}{2(2t-1)}\cdot\frac{t^2}{1+t^2} = \frac{dt^2}{2(2t-1)}.$$

The particle misses the roof exactly when $H < \frac{9}{10}d$. Since $2t - 1 > 0$ we may multiply through by $10(2t-1)/d$ without reversing the inequality:

$$\begin{aligned}
\frac{dt^2}{2(2t-1)} < \frac{9d}{10} &\iff 5t^2 < 9(2t-1)\\
&\iff 5t^2 - 18t + 9 < 0\\
&\iff (5t-3)(t-3) < 0\\
&\iff \tfrac{3}{5} < t < 3.
\end{aligned}$$

Every t in this interval satisfies $t > \frac{1}{2}$, so the earlier condition adds nothing. If $t = \frac{3}{5}$ or $t = 3$ the particle just touches the roof, which is excluded. Because the roof is never hit, the particle comes back down to the floor and lands inside the tunnel.

Finally, θ is between 0 and $\frac{\pi}{2}$, where tan is increasing, so

$$\arctan\tfrac{3}{5} < \theta < \arctan 3,$$

as required.

Checks

- At $t = 3$: $H = \dfrac{9d}{2 \cdot 5} = \frac{9}{10}d$, and at $t = \frac{3}{5}$: $H = \dfrac{(9/25)d}{2 \cdot (1/5)} = \frac{9}{10}d$. Both endpoints graze the roof, as they should.
- As $t \to \frac{1}{2}^+$ the required speed $u \to \infty$ and $H \to \infty$: the particle travels almost along the straight line to P, so it must be very fast and climbs very high afterwards.
- Numerically, with $d = 3$ and $g = 9.8$ we tested 10^5 random angles. For each we found u, confirmed that the path passes through P, and found that $H < \frac{9}{10}d$ exactly when $0.5404 < \theta < 1.2490$ (that is, $\arctan 0.6$ and $\arctan 3$ in radians). A step-by-step simulation also confirmed $H = u^2 \sin^2\theta/(2g)$.

Common slips

- Treating u as a separate unknown and getting stuck: it must be eliminated using the condition at P.
- Forgetting to check the sign of $2t - 1$ before multiplying the inequality through by it.
- Checking only that P is below the roof. The highest point of the path usually comes after P, so it is the greatest height that matters.

Beyond STEP

Writing the trajectory in terms of $\tan\theta$ turns projectile questions into quadratics in t. The same trick finds the "envelope" (bounding parabola) of all trajectories with a given speed, and gives the two possible angles for hitting a target. Problems like this, where some parameters are fixed by constraints and the rest form an allowed region, lead into optimisation and the calculus of variations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is an excellent next step for this style of mechanics.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.